Economics S5040: Dynamic Stochastic Modeling
Lecture 15 — Saving under Income Risk

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Fall 2026

Central question: How does a household smooth consumption when future income is uncertain and insurance markets are incomplete?

Plan for an 80-minute class

Minutes	Topic
0–8	Bridge from Week 7: saving becomes self-insurance
8–35	Two-period saving under uncertainty
35–48	Precautionary saving and the option value of assets
48–60	From two periods to the infinite-horizon household problem
60–72	Bellman equation, state variables, and borrowing constraint
72–80	VFI preview, interpretation, and summary

1 From Week 7 to Week 8

Last week the household moved consumption across dates. This week the household must also move resources across *states of the world*.

In a deterministic problem, saving trades consumption today for consumption tomorrow. Under income risk, saving has a second role:

assets provide self-insurance against bad future income realizations.

The household still wants smooth consumption, but future income is no longer known when today's saving decision is made.

2 A two-period example: saving before income is known

Consider a household that lives for two periods. At date 0 it has certain income y_0 and chooses saving a_1. At date 1 it receives uncertain labor income Y_1.

The budget constraints are

$$c_0 + a_1 = y_0, \qquad c_1 = Ra_1 + Y_1,$$

where $R = 1 + r$ is the gross return on saving. Initially impose

$$a_1 \geq 0,$$

so the household can save but cannot borrow against future labor income.

Preferences are

$$\max_{0 \leq a_1 < y_0} \{u(y_0 - a_1) + \beta \mathbb{E}\left[u(Ra_1 + Y_1)\right]\}.$$

This is already an income fluctuation problem in miniature. The household chooses a_1 *before* observing Y_1.

A simple two-state income risk

Let

$$Y_1 = \begin{cases} y_L & \text{with probability } p, \\ y_H & \text{with probability } 1 - p, \end{cases} \qquad y_L < y_H.$$

Then

$$U(a_1) = u(y_0 - a_1) + \beta \left[p\, u(Ra_1 + y_L) + (1 - p)u(Ra_1 + y_H)\right].$$

For an interior optimum,

$$-u'(c_0) + \beta R \left[p\, u'(c_1^L) + (1 - p)u'(c_1^H)\right] = 0,$$

so

$$\boxed{u'(c_0) = \beta R \mathbb{E}[u'(c_1)].}$$

Here

$$c_1^L = Ra_1 + y_L, \qquad c_1^H = Ra_1 + y_H.$$

Interpretation. One dollar not consumed today costs $u'(c_0)$ units of current marginal utility. Saving that dollar produces R dollars tomorrow, whose marginal utility depends on the income state. The household compares today's marginal utility with the *expected marginal utility* generated by saving.

3 A numerical classroom example

Take

$$y_0 = 10, \qquad \beta = 0.95, \qquad R = 1.02,$$

and log utility,

$$u(c) = \log c.$$

Suppose future income is

$$Y_1 = \begin{cases} 2 & \text{with probability } 1/2, \\ 10 & \text{with probability } 1/2. \end{cases}$$

Thus

$$\mathbb{E}[Y_1] = 6.$$

The household solves

$$\max_{0 \le a_1 < 10} \left\{ \log(10 - a_1) + \frac{0.95}{2} \log(1.02a_1 + 2) + \frac{0.95}{2} \log(1.02a_1 + 10) \right\}.$$

The first-order condition is

$$\frac{1}{10 - a_1} = 0.95(1.02) \left[\frac{1/2}{1.02a_1 + 2} + \frac{1/2}{1.02a_1 + 10} \right].$$

The comparison that matters

Now replace uncertain income by certain income equal to its mean:

$$Y_1 \equiv 6.$$

The deterministic problem is

$$\max_{0 \le a_1 < 10} \{ \log(10 - a_1) + 0.95 \log(1.02a_1 + 6) \}.$$

Ask the class before calculating:

> If expected future income is unchanged, should the household save the same amount?

With concave utility, bad states have especially high marginal utility. An additional dollar of assets is therefore particularly valuable when income turns out to be low. This creates a precautionary motive for saving.

> **Same mean, different risk.** Comparing risky income with certain income having the same mean isolates the effect of uncertainty on the saving decision.

For log utility,

$$\mathbb{E}[u'(Ra_1 + Y_1)] = \mathbb{E}\left[\frac{1}{Ra_1 + Y_1} \right].$$

Because $1/c$ is convex,

$$\mathbb{E}\left[\frac{1}{Ra_1 + Y_1} \right] > \frac{1}{Ra_1 + \mathbb{E}[Y_1]}$$

for nondegenerate risk. Hence, at a given a_1, risk raises expected future marginal utility and pushes toward greater saving.

4 What exactly is precautionary saving?

Consumption smoothing and precautionary saving are related but distinct ideas.

- **Consumption smoothing:** households prefer to avoid large fluctuations in consumption.
- **Self-insurance:** households use an asset buffer to transfer resources into bad income states.
- **Precautionary saving:** uncertainty itself can increase desired saving relative to an appropriate no-risk benchmark.

The asset is not state-contingent insurance. It pays R regardless of the income realization. Nevertheless, because the household chooses consumption after observing income, accumulated assets can be drawn down when income is low.

5 The borrowing constraint

Suppose the household would like to choose $a_1 < 0$. A borrowing constraint can prevent it:

$$a_1 \geq \underline{a}.$$

With the simple no-borrowing constraint, $\underline{a} = 0$.

The first-order condition must then be written as an inequality. At the optimum,

$$u'(c_0) \geq \beta R \mathbb{E}[u'(c_1)],$$

with equality when $a_1 > \underline{a}$.

Equivalently, introducing multiplier $\lambda \geq 0$,

$$u'(c_0) = \beta R \mathbb{E}[u'(c_1)] + \lambda,$$

$$\lambda(a_1 - \underline{a}) = 0.$$

If the constraint binds, the household would like to move still more consumption from the future into the present, but cannot.

6 From two periods to the income fluctuation problem

Now let the household live forever. A standard formulation is

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = R a_t + y(z_t), \qquad a_{t+1} \geq \underline{a},$$

where z_t follows a finite-state Markov chain

$$\Pr(z_{t+1} = z' \mid z_t = z) = P(z, z').$$

The state is now

$$\boxed{(a_t, z_t)}.$$

Why both variables? Assets summarize resources carried from the past, while z_t contains information about current income and, when income is persistent, the distribution of future income.

The control can be written as next period's assets a' (or, equivalently, current consumption). Since

$$c = Ra + y(z) - a',$$

the Bellman equation is

$$\boxed{V(a,z) = \max_{a' \geq \underline{a}} \left\{ u\big(Ra + y(z) - a'\big) + \beta \sum_{z'} P(z,z') V(a',z') \right\}.}$$

Feasibility also requires

$$Ra + y(z) - a' > 0.$$

7 The Euler equation and the constraint

At an interior solution,

$$u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})].$$

With a binding borrowing constraint,

$$u'(c_t) > \beta R \mathbb{E}_t[u'(c_{t+1})].$$

Thus the Euler *inequality* is economically meaningful: it identifies states in which the household is constrained.

8 Why persistent income requires another state variable

If income were IID, today's income would tell us nothing about tomorrow's distribution. With persistent income,

$$P(z_{t+1} \mid z_t)$$

depends on z_t. A household currently in a low-income state and one currently in a high-income state can therefore have different expectations about future income even when their asset holdings are identical.

This is why the income fluctuation problem naturally combines the tools from earlier weeks:

$$\boxed{\text{dynamic programming} + \text{Markov chains} + \text{optimal saving.}}$$

9 How we will compute the solution

For the first numerical solution we discretize the asset space,

$$\mathcal{A} = \{a_1, \ldots, a_N\},$$

and evaluate, for every current state (a_i, z_j) and feasible choice a'_k,

$$u(Ra_i + y(z_j) - a'_k) + \beta \sum_{z'} P(z_j, z')V(a'_k, z').$$

Value function iteration repeatedly applies this maximization until V converges.

This is not the fastest available method, but it is transparent and robust. In the next lecture we will exploit more structure to obtain faster methods.

End-of-class summary

By the end of today, students should be able to explain all four statements:

Income risk makes future marginal utility uncertain.
Assets allow households to self-insure.
Risk can create precautionary saving.
Borrowing constraints make the Euler equation an inequality in some states.

Reading

QuantEcon, *The Income Fluctuation Problem I: Discretization and VFI*:
`https://python.quantecon.org/ifp_discrete.html`